SS-W-110C
June 16, 1972
SUPERSEDING
Fed. Spec. SS-W-110b
February 7, 1963

FEDERAL SPECIFICATION

WATER-REPELLENT, COLORLESS, SILICONE RESIN BASE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a clear penetrating solution of silicone compounds which provide water repellency to exterior concrete, natural stone, brick, cast stone, unglazed terra cotta, and other porous masonry materials.

1.2 Classification. The water-repellent solution shall be a solvent soluble type.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-P-143 - Paint, Varnish, Lacquer and Related Materials; General Specifications for Packaging, Packing and Marking.
TT-T-291 - Thinner, Paint, Volatile Spirits (Petroleum-Spirits).

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Test Method Std. No. 158 - Cements, Hydraulic; Sampling, Inspection and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activites may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

C 144 - Aggregate for Masonry Mortar.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The water-repellent compound shall be a stable, clear, transparent, penetrating solution of silicone compounds in solvents free of suspended matter. When applied to clean, dry, concrete or masonry surfaces, it shall not cause preceptible change in color, surface texture or leave a visible film. It shall produce a water-repellent surface within 48 hours after application.

3.2 Composition. Shall consist of not less than a 5 percent solution by weight (total solids) of silicone resin when tested as prescribed in 4.3.1. The supplier shall, prior to commencing production, furnish to the government representative the identity and an infrared spectrographic curve of the silicone resin to be used in the finished product. The supplier may not change silicone resins for the duration of the contract.

3.3 Solvent. The solvent shall conform to TT-T-291, type II, grade A or any solvent system complying with the following requirements by volume:

(a) Aromatic compounds with eight or more carbon atoms except ethyl benzene: 8 percent maximum.
(b) Ethyl benzene and toluene: 20 percent maximum.
(c) Solvents or thinners with an olefinic or cycle-olefinic type of unsaturation: negative test.
(d) Ketones: negative test.
(e) Total of (a) plus (b): 20 percent maximum.

A certificate of compliance from the supplier to this effect is necessary.

3.4 Storage stability. The test specimens when stored for at least six months at 0° to 100°F. shall show no signs of separation, settling, solidification and shall meet all the requirements of the specification.

3.5 Water repellence. The test specimens shall have an average water absorption of less than 1 percent by weight when tested in accordance with 4.3.2.

3.6 Weathering. The test specimens shall show an average water absorption of less than 2 percent by weight when tested for weathering in accordance with 4.3.3.

3.7 Breathing. The test specimens when tested in accordance with 4.3.4 shall have a moisture loss of not less than 50 percent.

3.8 Resistance to efflorescence. The solution shall show no visible efflorescence when tested in accordance with 4.3.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling and inspection shall be in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.2.1 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table I. For examination of exterior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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TABLE I. Classification of preparation for delivery defects

Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Quantity per container is more or less than required. Net weight exceeds requirements.

4.3 Test procedures. Selected samples shall be tested in accordance with the applicable methods of Fed. Test Method Std. No. 141 and as specified herein. Failure to pass any test shall be cause for rejection of the lot.

4.3.1 Total solids. The material shall be tested in accordance with method 4041 of Fed. Test Method Std. No. 141 except that the dish and its contents are heated over a steam bath until the bulk of the volatile matter has evaporated, prior to drying in the oven.

4.3.1.1 Silicone resin content. A film is prepared on a NaCl plate by evaporating the solvent from a sample with gentle heating. The thickness is adjusted to give an absorbance value of 1 to 1.5 for a major band. Compare the spectrum with a standard spectrum in an infrared spectrophotometer in accordance with instructions supplied by the instrument manufacturer. A positive qualitative identification is made if the unknown material spectrum matches that of the known material for number, positions, shapes, and relative intensity of the bands.

4.3.2 Water repellence. Mortar cubes to be used as test specimens shall be prepared in accordance with method 2301.1 (and certain applicable paragraphs of method 2001.1 as are indicated) of Fed. Test method Std. No. 158, except as follows:

(a) Sand conforming to ASTM C 144 may be used instead of the Ottawa sand specified.
(b) The proportions of water, cement, and sand shall be adjusted to produce at the end of 28 days (1) a compressive strength of not less than 1000 pounds per square inch and (2) an absorption of not less than 8 percent based on the dry weight of the mortar.

4.3.2.1 Compressive strength of test specimens shall be determined in accordance with method 2301.1 of Fed. Test Method Std. No. 158.

4.3.2.2 Absorption. Test specimens having the required compressive strength shall be dried at a temperature of 80°C., plus or minus 5°C. until constant weight has been reached, and allowed to cool to room temperature. The specimens shall then be weighed to the nearest 0.1 g. and placed with the cut-off surface upward in 1/4 inch constant head of water for 24 hours. At the end of this time, remove the cubes from the water, remove the excess surface water with a damp cloth, weigh at once and calculate the amount of water absorbed as a percentage of the dry weight of the cube.

The cubes shall then be dried again to constant weight and used at once for the water repellence test.

Constant weight shall be assumed when the specimen weighed to the nearest 0.1 gram loses not more than 0.2 gram in any four-hour period.

The cubes shall be totally immersed in the water repellent solution under test for ten seconds. Three cubes shall be run for each water repellent solution. The treated cubes shall be allowed to cure for 48 hours at 20° to 24°C. at a relative humidity of 20 ± 5 percent. At the end of this time the cubes shall be weighed to the nearest 0.1 gram and immersed in 1/4 inch depth of water with cut-off side upward for 72 hours.

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The cubes shall then be removed from the water, blotted with a damp cloth to remove excess water, and weighed. The percent water absorbed by the treated specimen is then calculated based on the weight of the dry treated cube.

4.3.3 Weathering. Any three of the mortar cubes, prepared and coated with water repellent solution and found to meet the water absorption test (as specified above under 4.3.2) shall be exposed in a weatherometer for 300 hours in accordance with Fed. Test Method Std. No. 141, method 6151. After exposure, the three mortar cubes shall be again tested for water absorption as specified in 4.3.2.2.

4.3.4 Breathing. Grade M common brick, or other suitable brick, capable of absorbing 10 to 20 percent of its dry weight of water shall be used in this test. The brick shall be dried in a temperature of 100°C., ± 5°C., to constant weight (4.3.2.2) and its weight recorded, allowed to cool and then treated on its four edges and one bedding surface with one brush flooded coat of the water repellent solution under test. The treated brick shall be allowed to cure for 48 hours at 20 to 24°C. and then completely immersed with its untreated surface downward in water. The brick shall be supported to permit free circulation of water around all surfaces and edges. It shall remain in water storage until it is completely saturated as determined by constant weight (4.3.2.2), removing excess surface water with a damp cloth, and its saturated weight recorded. Total water absorption shall then be determined and recorded.

The saturated brick shall be placed at once, untreated surface downward, on an oversized piece of waxed paper and the paper sealed to the brick by pouring a hot sealing wax fillet between the paper and the vertical edges of the brick. The prepared specimen shall be kept at a temperature of 20 to 24°C., at a relative humidity of 50 ± 5 percent with a free circulation of air for seven days. At the end of this period, the moisture loss shall be determined and expressed as a percentage of the total water absorbed.

4.3.5 Resistance to efflorescence. Remove the wax and waxed paper from the brick used in 4.3.4 and again dry the brick to constant weight. (4.3.2.2). Then place the brick with the untreated surface downward in 1/2 of 10 percent sodium sulphate aqueous solution maintained for seven days at room temperature, then remove the brick and dry for 24 hours in a ventilated oven and examine for visible efflorescence.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The water-repellent shall be packaged, packed, and marked in accordance with TT-P-143. The levels of packaging and packing shall be A, B, or C as specified (see 6.2). The marking shall be civil or military as specified (see 6.2). Quantities shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. To provide protection against the entrance of rain water and the moisture from melting snow, ice, or frost into the pores of exterior concrete and masonry surfaces above grade, including tight mortar joints. It is not intended as a waterproofing coating. Hard driving rain can penetrate into and through the pores of silicone-resin treated concrete and masonry surfaces.

6.1.1 It is intended to provide water-repellence in one application and yet permit any water within the concrete or masonry to escape by evaporation.

6.1.2 When properly applied to clean, dry surfaces, it should penetrate to an appreciable depth, depending on the character of the materials treated, without leaving a visible surface film and without causing perceptible change in color, texture, and appearance.

6.1.3 It is not intended to seal visible cracks and crevices or as a substitute for repointing defective mortar joints.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurements documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging, packing and marking required (see 5.1).
(c) Specify whether civil or military marking required (see 5.1).

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6.3 Condition of concrete and masonry surfaces. The surface on which the water repellent is to be applied should be clean and dry. Existing dirt, grease, asphalt, tar, stains or resinous materials should be removed from such surfaces by approved methods prior to application of the water repellent. Similarly, all other structural defects such as poor flashing, poor caulking, etc. should be repaired before the water-repellent is applied.

The structure should be allowed at least three bright, warm days to dry prior to applications of the water repellent, and the water repellent applied on a day when no rain is expected for 12 hours after application.

All visible cracks or crevices due to the shrinkage or settlement of concrete or the breaking or displacement of masonry facing, and all mortar joints that are not tight and sound should be repaired by approved methods prior to the application of the water repellent.

6.3.1 Coverage. The water repellent should be applied by brush or light spray, flooded on the surface and not be brushed out. If spray technique is used, the materials should be forced from the gun in an easy stream and not atomized or misted. A pressure of not more than 5 pounds at the applicator's spray is recommended. The gun shall be held close to the wall to provide a wet application.

Application should be such that a run down of material on the surface of six inches to one foot occurs during application. Coverage shall be on the order of 75 to 100 square feet per gallon applied in one coat for most surfaces. Dense materials which cannot take one gallon per 100 square feet in one coat, should be given two applications of 150 to 200 square feet per gallon each coat.

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 80

Military Coordinating Activity:

Army - ME

CIVIL AGENCIES INTEREST:

COMMERCE - NBS
USDA - AFS

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